

Harbourline Energy Co.

Lockout and Tagout Safety Manual

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Applies to	All employees and contractors who isolate, work on, or work near electrical and mechanical equipment owned or operated by Harbourline

1. Purpose

This manual sets the minimum requirements for isolating energy sources, applying locks and tags, and verifying a zero energy state before work, so that no worker is exposed to the unexpected energization or release of stored energy. It also sets Harbourline's minimum approach distances to exposed energized parts.

2. Scope and Regulatory Basis

The manual applies to all generation, transmission, substation, distribution and facility equipment, including mechanical, hydraulic, pneumatic and thermal energy sources. It implements the Ontario Occupational Health and Safety Act and its regulations for electrical utility work in Ontario, and OSHA 29 CFR 1910.147 and 1910.269 in New York and Ohio. Where these differ, the more protective requirement applies in all three jurisdictions.

3. Definitions

- **Authorized Operator:** a qualified person designated by the Distribution System Operator or station operations to perform isolation switching and issue permits.
- **Permit Holder:** the qualified worker in charge of the work who accepts the permit and holds it until the work is complete.
- **Isolation point:** a device that physically prevents the transmission or release of energy, such as a disconnect switch, breaker racked out, valve, or removed fuse.
- **Zero energy state:** the condition in which all energy sources are isolated, stored energy is released or restrained, and absence of voltage is verified.
- **Working grounds:** temporary protective grounds applied to create an equipotential zone at the work location.

4. Responsibilities

Role	Responsibilities
Authorized Operator	Prepares the isolation list, performs switching, applies operations locks and tags, and issues the permit.
Permit Holder	Verifies isolation, applies working grounds, briefs the crew, controls the group lockbox, and surrenders the permit.
Each worker	Applies a personal lock to the group lockbox before starting work and removes only their own lock.
Supervisor	Ensures workers are trained, audits permits, and authorizes absent worker lock removal under Section 11.
Health and Safety	Maintains this manual, the training program, and the annual audit schedule.

5. Lockout Devices and Tags

Device	Colour	Who applies it
Operations lock	Yellow	Authorized Operator, at each isolation point
Personal lock	Red	Each worker, on the group lockbox or directly at the isolation point
Group lockbox	Blue	Permit Holder, holds the keys to all operations locks
Contractor personal lock	Orange	Contractor workers, issued at site orientation
Danger tag	Red and white	Attached to every operations lock and personal lock

Each personal lock is keyed individually, marked with the worker's name and employee number, and has one key held only by that worker. Tags must show the permit number, the name of the person who applied it, the date and time, and the reason for isolation.

6. Permit System

6.1 Permit numbering format

Every lockout is controlled by a Work Protection Permit. Permit numbers use the format LOTO-RRR-YYYY-NNNNN, where RRR is the three letter region code, YYYY is the year of issue, and NNNNN is a five digit sequence number that restarts at 00001 on January 1 in each region. Example: LOTO-EON-2026-00417 is the 417th permit issued in Eastern Ontario in 2026.

Region code	Region
EON	Eastern Ontario
CON	Central Ontario
DUR	Durham
NYN	New York North Country
NYW	Western New York

Region code	Region
OHN	Northeast Ohio
OHC	North Central Ohio
TRN	Transmission (all jurisdictions)

6.2 Permit validity

- A permit is valid until the work is complete and the permit is surrendered, up to a maximum of 7 calendar days. Work lasting longer requires a new permit with a new number.
- A permit may be transferred at shift change only under Section 9.
- Permit records are kept in the Work Protection register for 7 years.

7. Lockout Procedure

Step 1. Prepare: the Authorized Operator identifies every energy source using the single line diagram and the equipment isolation list, and records each isolation point on the permit.

Step 2. Notify: inform the DSO or station operator and all affected workers that the equipment will be isolated.

Step 3. Shut down: stop the equipment using normal operating controls.

Step 4. Isolate: open every isolation point. Rack out breakers, open disconnects and remove fuses where applicable.

Step 5. Lock and tag: apply a yellow operations lock and danger tag to each isolation point and place all operations lock keys in the blue group lockbox.

Step 6. Issue the permit: the Authorized Operator and the Permit Holder review the permit together and both sign it.

Step 7. Release stored energy: discharge capacitors, bleed hydraulic and pneumatic pressure, block springs and raised parts.

Step 8. Verify absence of voltage: test with a rated voltage detector, proving the detector on a known live source before and after the test.

Step 9. Apply working grounds: install temporary protective grounds on all sides of the work location.

Step 10. Apply personal locks: every worker applies a red personal lock to the group lockbox and signs the permit before starting work.

7.1 Restoring equipment

When work is complete, each worker removes their own personal lock and signs off. The Permit Holder confirms tools and personnel are clear, removes working grounds, and surrenders the permit to the Authorized Operator, who removes the operations locks and returns the equipment to service.

8. Group Lockout

For crews of more than one worker, the group lockbox method is mandatory. The Permit Holder is the first to apply and the last to remove a personal lock. Contractors apply orange personal locks to the same group lockbox and are listed by name on the permit.

9. Shift Change Transfer

The outgoing and incoming Permit Holders review the isolation points and grounds in person or by recorded phone call with the Authorized Operator. The incoming Permit Holder applies a personal lock before the outgoing Permit Holder removes theirs, so the lockbox is never without a Permit Holder lock.

10. Minimum Approach Distances

No person may approach, or take any conductive object closer than, the distances below to exposed energized parts unless the part is isolated and grounded under this manual, or the worker is a qualified worker using approved insulated tools and cover up. Harbourline company values meet or exceed the applicable regulations. Voltages are nominal phase to phase.

Nominal voltage (kV)	Qualified worker MAD	Unqualified person MAD
0.050 to 0.300	Avoid contact	Avoid contact
0.301 to 0.750	0.35 m (1 ft 2 in)	1.0 m (3 ft 4 in)
0.751 to 15.0	0.70 m (2 ft 4 in)	3.0 m (10 ft 0 in)
15.1 to 36.0	0.85 m (2 ft 10 in)	3.0 m (10 ft 0 in)
36.1 to 46.0	1.00 m (3 ft 4 in)	3.0 m (10 ft 0 in)
46.1 to 72.5	1.20 m (4 ft 0 in)	3.3 m (10 ft 10 in)
72.6 to 121	1.40 m (4 ft 7 in)	3.8 m (12 ft 6 in)
138 to 145	1.60 m (5 ft 3 in)	4.0 m (13 ft 2 in)
161 to 169	1.80 m (5 ft 11 in)	4.2 m (13 ft 9 in)
230 to 242	2.40 m (7 ft 11 in)	4.9 m (16 ft 1 in)
345 to 362	3.80 m (12 ft 6 in)	6.0 m (19 ft 8 in)
500 to 550	5.60 m (18 ft 5 in)	7.6 m (25 ft 0 in)

Common Harbourline distribution voltages: 4.16 kV, 8.32 kV, 12.47 kV (Ohio), 13.2 kV (New York), 13.8 kV, 27.6 kV and 44 kV (Ontario). Common transmission voltages: 115 kV and 230 kV.

11. Removal of a Lock by Someone Other Than the Owner

Step 1. The supervisor makes at least three documented attempts to contact the lock owner, including a call to their personal phone.

Step 2. The supervisor confirms the owner is not at the work site.

Step 3. The Regional Operations Manager approves the removal in writing on the permit.

Step 4. The lock is cut in the presence of a second qualified worker.

Step 5. The owner is informed before returning to work and signs the permit record.

12. Training and Audits

- Initial training before first isolation duty, and refresher training every 12 months.
- Supervisors audit at least 2 permits per crew per quarter in the field.
- Health and Safety audits each region's Work Protection register annually.

- Any lockout failure or near miss is reported as a safety incident within 24 hours.

Revision History

Revision	Date	Author	Summary of changes
7	2022-01-10	A. Coombs	Added contractor orange personal locks.
8	2024-02-01	A. Coombs	Permit numbering changed to include three letter region code.
9	2026-01-15	A. Coombs	Added Durham (DUR) region code; updated minimum approach distance table.